

# Automated Minutiae Extraction and Fingerprint Analysis using Machine Learning Techniques

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**Abstract-** In the digital world, traditional methods of biometric recognition are being challenged. Traditional systems such as RFID and cards are vulnerable to fraud. However, fingerprints contain varied degrees of unique identifiers for each individual and can be exploited for precise recognition of people. Automatic feature extraction, a key part of fingerprint analysis, identifies unique ridge and edge patterns in a fingerprint image. This information can be digitally signed to enable secure identification and verification, eliminating the risk of manipulation. This work proposes a fingerprint analysis and matching system using deep learning and machine learning techniques. For feature extraction, edge detection and adaptive thresholding have been implemented. To prevent spoofing attacks, templates are generated using pattern recognition (using level 1 features fingerprints) and minutiae (also known as level 2 feature) of fingerprints. The work compares CNN, Random Forest, KNN, and SVM for fingerprint matching. The analysis of ML models reveals that CNN performs better compared to others with an accuracy of 97.6%.

**Keywords:** Minutiae, CNN, KNN, Random Forest, SVM, Edge Detection, Adaptive Thresholding

## I. Introduction

Our everyday lives are increasingly dependent on secure and reliable identification

methods. Traditional attendance systems, often relying on cards or codes, are vulnerable to loss, sharing, or even theft. Fingerprint recognition technology offers an effective solution to these challenges. Fingerprints are a unique identifier for each individual, and they remain relatively unchanged throughout life. This uniqueness makes them highly suitable for secure attendance systems. The fingerprint recognition technology captures the intricate network of ridges and valleys on a person's fingertip. These distinctive patterns are subsequently translated into a digital fingerprint, which is essentially a mapped representation of the ridges and valleys. This digital representation serves as a secure and tamper-proof identification for verification purposes. Fingerprint recognition has several advantages over other biometric identification methods. Fingerprint scanners are more compact and can be easily integrated into everyday devices. While voice recognition is convenient, it can be compromised by background noise or accents. Fingerprints, however, remain a consistent and distinct identifier, presenting a high level of precision and security. Modern fingerprint recognition systems employ sophisticated algorithms and high-resolution scanners to achieve exceptional accuracy and speed. Prior limitations, such as difficulties with dry, damaged, or scarred fingerprints, have been addressed. Today's systems can effectively capture and analyse these variations, ensuring a smooth and inclusive user experience.

The applications of fingerprint recognition

technology extend far beyond the basic identification function. It can be used for secure access control in buildings or restricted areas, for secure online transactions and mobile payments, and even for age verification. Fingerprint recognition facilitates a more secure environment by eliminating the possibility of unauthorised access or identity theft. The rise of deep learning has revolutionized the field of computer vision, tackling problems once deemed insurmountable for traditional machine learning algorithms [1], [2]. Convolutional Neural Networks (CNNs) have been particularly transformative, offering efficient solutions to image processing challenges. This has significantly impacted fingerprint recognition. CNN-based fingerprint recognition systems have surged in popularity due to their exceptional ability to extract and analyse the intricate details of fingerprints, leading to highly accurate and secure identification.

### I.1 Fingerprint Analysis Techniques

1) *Minutiae-Based Recognition - The Gold Standard*: Minutiae-based recognition stands as the most widely employed technique, garnering significant attention in research. This approach focuses on the distinctive ridge endings and bifurcations present within a fingerprint image. The underlying principle is to extract and match the minutiae between a captured fingerprint and a pre-enrolled template stored in the database. The intricate details captured by minutiae offer a high degree of distinctiveness, translating into superior accuracy and reliability in fingerprint matching.

2) *Correlation-Based Recognition*: Correlation-based recognition provides a computationally efficient approach to fingerprint verification. This technique involves directly comparing the entire fingerprint image captured with a pre-existing template, thereby generating a correlation score to evaluate their similarity. This method offers simplicity and ease of implementation, but its performance can be susceptible to noise and distortions present in fingerprint images [6].

3) *Pattern-based recognition*: The concept of pattern-based recognition focuses on the classification of fingerprints based on broad fingerprint patterns, such as arches, loops, and whorls. This approach proves valuable for initial classification tasks, particularly in large fingerprint databases. Nonetheless, owing to the inherent

limitations in the level of detail captured by fingerprint patterns, this technique offers less distinctiveness in comparison to minutiae-based methods for verification purposes.

4) *Hybrid recognition*: Hybrid recognition techniques combine the strengths of several approaches to achieve enhanced accuracy and robustness in fingerprint recognition. This frequently involves the amalgamation of minutiae analysis with other methodologies, such as pattern classification or correlation matching, as demonstrated in [7], [8]. By leveraging the complementary information captured by each method, hybrid recognition offers a powerful solution for high-security applications that require exceptional accuracy and reliability.

## II. Literature Review

Building on the promising application of deep learning for fingerprint minutiae extraction demonstrated by Darlow et al. [1], the study utilizes this technique to further explore its potential in fingerprint recognition systems. Their work established a foundation for utilizing deep neural network architectures for this task. Recognizing the *advantages* of this *approach*, including its potential for high *precision* and *efficacy*, the study utilizes CNN to replicate and evaluate its effectiveness. While neural networks have shown promise in fingerprint extraction, existing methods often suffer from slow processing times. Zhou et al. [2] introduce a novel two-stage neural network architecture that addresses this limitation. The first stage identifies candidate regions containing minutiae, while the second stage identifies individual pixels within those regions. This method achieves significantly faster processing times compared to previous neural network methods on public datasets (FVC 2002 & 2004) without sacrificing accuracy. This work demonstrates the potential of neural networks for efficient and accurate fingerprint minutiae extraction.

Takahashi et al. [10] discuss a novel method for fingerprint feature extraction using a Multi-Task Convolutional Neural Network (CNN). This method combines texture, minutiae, and frequency spectrum for enhanced feature extraction. It introduces a minutia attention module and new data augmentation methods. The proposed method outperformed commercial

fingerprint-matching software and the conventional method. Zhang et al. [11] present a CNN that is capable of performing multiple tasks simultaneously, namely the detection of minute points and the computation of their directions in contactless fingerprint images. This approach is notable for its direct operation on raw grayscale images without the need for preprocessing. It incorporates an attention mechanism to enhance the direction regression task, focusing specifically on the minute details.

The study presented by Durú et al. [12] provides a comprehensive evaluation of the efficacy of various classifiers in biometric recognition based on fingerprints. The study reveals that all classifiers achieve commendable recognition rates, with SVM and k-NN standing out due to their robustness against parameter variations. The quality of fingerprint images has become a pivotal factor influencing classifier performance. The research demonstrates the potential of SVMs with alternative kernel functions and the synergistic effect of hyper-parameters on performance enhancement. It advocates for the exploration of the collective impact of diverse vector machine models in the future, intending to elevate accuracy and generalization capabilities beyond the accomplishments of individual classifiers. Prasad et al. [13] present a comprehensive assessment of the current state of fingerprint recognition. The paper examines the intricacies of biometric authentication, emphasising the challenges in designing systems that accurately extract and match fingerprint features, especially under suboptimal conditions. It criticizes the misconception that fingerprint recognition is a fully matured field, instead highlighting the ongoing complexities in pattern recognition for authentication. It describes various preprocessing techniques, such as binarization and false minutiae elimination, to improve the quality of minutiae points used for authentication. They also reviewed algorithms for minutiae matching and their performance, contributing to understanding biometric authentication potential and limitations.

The study by Khard et al. [14] stresses the importance of biometric authentication, especially fingerprint recognition, in enhancing security. It examines various approaches to improve accuracy and performance, including image denoise methods. They also explore cross-ridge frequency analysis and statistical measures to calculate the

weighted mean phase. The study utilizes a fuzzy c-means classifier to assess features such as ridge reliability or ridge centre frequency. This provides insights into the vulnerability of fingerprint systems to spoofing attacks. The work by Guan et al. [16] presents a comprehensive analysis of the preprocessing of latent fingerprint images, which are crucial for forensic analysis, yet frequently exhibit poor quality from crime scenes. The article outlines the establishment of a novel database containing latent fingerprint images and the implementation of a study wherein trained Latent Print Examiners (LPEs) provided Extended Feature Set (EFS) markups for these images. The study's objective was to quantitatively evaluate the significance of latent fingerprint images before and after preprocessing, with a particular emphasis on the variations in detectable fingerprint minutiae and quality scale variations. The study revealed that preprocessing significantly enhances the quality and quantity of fingerprint information, which is crucial for conducting evidentiary analysis. The paper also discusses the experimental design, the metrics used to measure the increase in fingerprint information, and the analysis of these metrics. It is concluded that preprocessing transforms latent images of marginal value into ones suitable for analysis, indicating that colour images may possess significant value for Automated Fingerprint Identification Systems (AFIS). The study aims to provide a scientific basis for latent fingerprint analysis and contribute to the development of comparable analysis methods for other image-based methods in forensic science.

Chaurasia et al. [17] delves into the realm of biometric identification through fingerprints, underscoring the importance of preprocessing in enhancing the accuracy of minutiae detection and feature extraction. The paper meticulously evaluates existing algorithms and proposes a specific sequence of processing steps that significantly enhances the quality of the final output, provided that the initial image quality is sufficient. Its methodological approach assumes the use of high-quality imaging devices, thereby bypassing the need for image quality enhancement techniques. This assumption serves as both a limitation and a strength, as it streamlines the preprocessing phase, yet heavily relies on the initial image capture quality. The study demonstrates that the order in which algorithms are applied to a fingerprint image can significantly affect the effectiveness of feature extraction. The

paper discusses a series of steps, including binarization, central line thinning, dilation, and refinement, which collectively facilitate the production of a dual image that is primed for minutiae detection. The research offers a valuable perspective on the preprocessing of fingerprint images. It highlights the critical role of algorithmic sequencing in developing robust biometric recognition systems.

Sokoto Coventry Fingerprint Dataset (SOCOFing)[15] is a unique biometric dataset that provides a rich source of fingerprint images for academic research. It was developed by Shehu et al. and comprises 6,000 images from 600 African subjects, each labelled with gender, hand, and finger information. The dataset is further enhanced with synthetic alterations representing common distortions in fingerprint recognition systems, such as obliteration, central rotation, and z-cut. These alterations represent common distortions in fingerprint recognition systems, such as obliteration, central rotation, and z-cut. This enables researchers to test the resilience of fingerprint analysis algorithms under varied conditions. SOCOFing's focus on African demographics fills a critical gap in biometric research by offering a dataset that reflects the diversity of global populations. It is an invaluable tool for developing more robust and inclusive fingerprint recognition technologies. SOCOFing encourages innovation and collaboration in the field, supporting advances in pattern recognition and computer vision.

### **III. Minutiae Extraction and Fingerprint Analysis**

#### **III.I Data Preprocessing**

This study makes use of the SOCOFing Dataset [15], which comprises fingerprints from 600 African subjects. It takes into account the real images and augmented images. Augmented images provide more insights into learning and improve training by bringing improved precision. The dataset consists of high-quality, open-sourced images making it a better choice when used with machine learning models. The application involves data acquisition, image preprocessing and enhancement, feature extraction and template

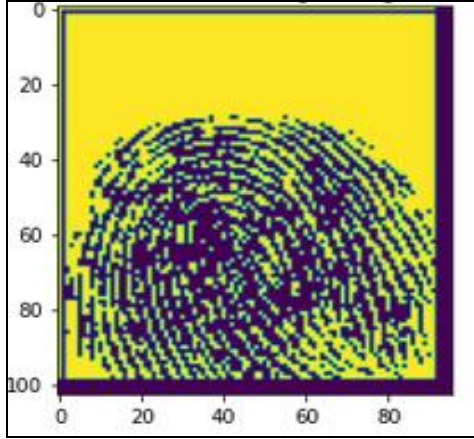
generation, model building with different machine learning models and matching. Here data acquisition involves utilising the massive SOCOFing Dataset and coming up with matching techniques for both subject identification and finger number identification.

Preprocessing of fingerprint images is an essential step in fingerprint-based biometric systems. Grayscale is a common preprocessing technique used in fingerprint biometrics. Fingerprint images are usually captured in colour. However, for many fingerprint feature extraction algorithms, the colour information is redundant and can introduce unnecessary complexity. Converting an image to grayscale reduces data dimensionality and simplifies subsequent processing steps without significant information loss. Following grayscale, fingerprint image enhancement techniques further improve the image data quality for feature extraction. Edge detection algorithms possess significant value in highlighting fingerprint ridges, which are the raised lines that form the distinctive patterns used for identification. By enhancing these ridges, feature extraction algorithms can more effectively locate and characterise the fingerprint's minutiae, the ridge ends and bifurcations that provide the most distinctive fingerprint features.

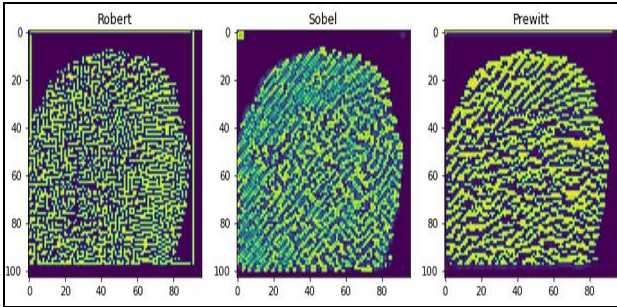
#### **III.II Minutiae Extraction**

Adaptive thresholding can separate the foreground fingerprint region from the background. This technique generates a binary image wherein the foreground pixels are assigned a specific value, commonly white, while the background pixels are assigned a distinct value, typically black. This segmentation step facilitates feature extraction by solely focusing on the relevant fingerprint region. Incorporating these preprocessing and enhancement techniques enhances the quality of fingerprint images, resulting in more robust and precise feature extraction for biometric applications. Otsu thresholding is an automatic technique used to separate foreground from background in grayscale images. The algorithm scrutinizes the histogram of the image to determine the intensity level that most effectively divides the pixels into two distinct classes. Otsu's method ensures that there is a clear distinction between foreground and background in the resulting binary image. This technique proves

to be particularly efficacious for images exhibiting a bimodal histogram, wherein the foreground and background exhibit distinct intensity peaks. However, it may encounter difficulties with images with noisy or complex intensity distributions.



**Fig. 1.** Otsu's thresholding

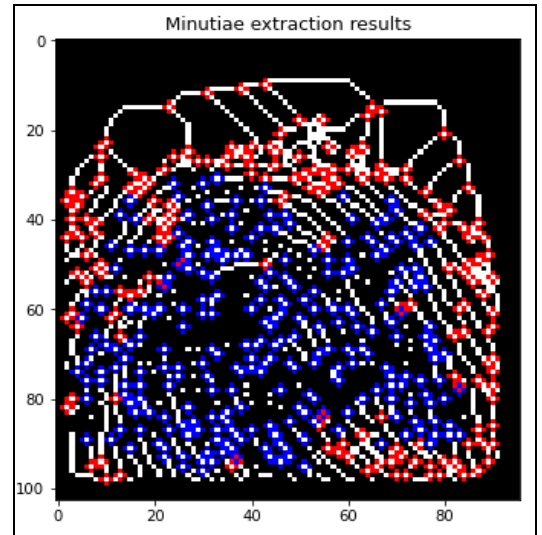


**Fig. 2.** Edge detection using Robert, Sobel, and Prewitt filters

Fingerprint imaging relies heavily on the accurate identification of ridge structures. These unique, winding patterns are the key to fingerprint recognition. To enhance these ridges and facilitate subsequent feature extraction, edge detection techniques play a crucial role. This work utilises gradient-based filters, specifically Roberts, Sobel, and Prewitt filters, for the enhancement of fingerprint ridges. These filters operate by calculating an approximate derivative of the image intensity in local neighbourhoods. This derivative highlights regions with rapid intensity changes, which often correspond to edges of fingerprint ridges. The Roberts filter utilizes a simple 2x2 kernel to estimate diagonal derivatives. The Sobel and Prewitt filters, however, employ three-

dimensional kernels that capture horizontal and vertical image gradients. These filters provide a more thorough analysis of local intensity variations than the Robert filter. By applying these filters to the fingerprint image, we obtain a filtered image that emphasizes ridge structures as shown in Figure 2. The magnitude of the filter response indicates the degree of edge strength, with higher values indicating more prominent ridges. This enhanced image data is then suitable for subsequent feature extraction algorithms that can accurately locate and characterize the minutiae (ridge ends and bifurcations) within the fingerprint patterns. The selection of an optimal filter among Roberts, Sobel, and Prewitt depends on factors such as computational efficiency and noise sensitivity.

Figure 3 describes extracting minutiae from fingerprint recognition systems, the critical features that are essential to fingerprint recognition systems. The approach effectively identifies and characterizes ridge ends and bifurcations within fingerprint images. These extracted minutiae provide a concise and dependable representation that is suitable for fingerprint matching and identification tasks. The algorithm is based on a multi-stage procedure. Initially, it performs the detection of termination and bifurcation. The image is analyzed by examining small, localized neighbourhoods (typically 3x3).



**Fig. 3.** Minutiae extracted from a fingerprint

Terminations are identified when a single white pixel is surrounded by black pixels within a block. Bifurcations, where a ridge splits, are detected when a block contains three white pixels

arranged in a specific configuration.

The algorithm then determines the orientation of each detected minute point. This step is critical to achieving rotation invariance, which ensures accurate matching regardless of the fingerprint's orientation during capture. The estimation of the orientation is derived by analyzing the relative positions of adjacent pixels within the local block surrounding each minutia point. Following detection and orientation estimation, the algorithm extracts feature from identified minutiae. Each minutia is represented by a separate Minutiae Feature object. This object contains essential information about the minutiae point, including its location (x and y coordinates), orientation, and type.

The minutiae extraction is performed on both altered and real datasets of the SOCOFing dataset. The features are used in the models built using CNN, KNN, SVM and Random Forest. For the CNN, two models are built to identify the subject and the finger used. To minimize overfitting, several regularization techniques were employed within these models. These techniques included L2 regularization, batch normalization, dropout regularization, early stacking, and reduce LR on plateau. By incorporating these methods into the models, the models were trained to generalize better and avoid overfitting the training data.

### III.III Training and analysis of models

The model architecture utilizes two separate convolutional neural networks (CNNs) designed for different tasks: identifying the subject ID (SubjectID\_Mod) and determining the finger number (FingerNum\_Mod). Both models exhibit a resembling structure. They begin with a convolutional layer with 32 filters of size 5x5 applied to the extracted features. This layer captures local features within the image using a corrected linear unit activation function. Batch normalization is then applied to improve training stability. A maximum pooling layer follows, downsampling the feature maps by a factor of two in both width and height. This improves the model's complexity and computational cost. Two additional convolutional layers with 64 and 128 filters are employed, followed by batch normalization and maximum pooling. These layers

remove increasingly complex features from the fingerprint image. A dropout rate of 0.3 is then introduced to prevent overfitting by randomly dropping out portions of neurons during training. The model then flattens the feature maps into a one-dimensional vector, which is then fed into two fully connected layers. The first dense layer has 256 neurons with a ReLU activation rate, followed by a dropout rate of 0.4 for further regularization. The final output layer has many units depending on the specific task, 600 for SubjectID\_Mod and 10 for FingerNum\_Mod, and uses a softmax activation for multi-class classification. Finally, the model is compiled using the Adam optimizer with a learning rate of 0.0001 and a categorical cross-entropy loss function. Accuracy is the primary metric used to track training progress.

Fingerprint classification using a KNN relies on features extracted from fingerprint images. These characteristics encompass specific details such as ridge terminations, bifurcations, local ridge orientations, or even frequency domain information. During the classification process, features are extracted from an unknown fingerprint image. The KNN algorithm then calculates the chamfer distance between the unknown fingerprint features and all of the features in the training data. Using these distances, the algorithm determines the k nearest neighbours (fingerprints) within the training data. Finally, a predicted class label for the unknown fingerprint is assigned based on the most frequent class label among its k closest neighbours.

Utilisation of SVM in fingerprint classification involves special attention to the kernel function. The linear kernel used in the code may not be optimal. Fingerprint data frequently reaps the benefits of non-linear kernel functions, such as the radial basis function (RBF kernel). These kernels enable the SVM to acquire more intricate decision boundaries in the feature space, thereby facilitating a more effective separation between diverse fingerprint classes. The training phase involves the training of an SVM classifier equipped with an RBF kernel using a dataset comprising labelled fingerprint features and their corresponding class labels class that could represent a unique fingerprint belonging to a specific individual. During classification, features are extracted from an unknown fingerprint image. The trained SVM classifier then uses the RBF kernel to classify the unknown fingerprint based

on the learned decision boundary. By incorporating a non-linear kernel function, like the RBF kernel, SVM can achieve better classification accuracy for fingerprint data compared to the linear kernel.

The random forest methodology relies on decision trees, and the construction of these trees necessitates consideration of the distinct characteristics of fingerprint data. The characteristics employed to partition data at each node in the trees ought to be pertinent to fingerprint classification. A random forest is constructed from a dataset containing labelled fingerprint features and their corresponding class labels. Each tree in the forest is constructed by splitting the data at each node based on fingerprint-specific features. This ultimately leads to a diverse ensemble of decision trees. Hyperparameters such as the number of trees and tree depth are optimized for the particular dataset. During classification, features are extracted from an unidentified fingerprint image. Each tree in the random forest makes a classification prediction for an unknown fingerprint based on the learned decision rules. The final classification label for the unidentified fingerprint is determined by a majority vote based on the predictions from all trees in the forest. Random forest leverages the efficacy of multiple decision trees, thereby delivering robustness and potentially higher classification accuracy for fingerprint data in comparison to a singular decision tree or the conventional KNN approach.

The application is developed using Qt, making it a suitable option to be used as an embedded systems application. The models are built using CNN for person identification and finger number recognition.

#### IV. Results

The dataset was split into train, validation and test sets for analysis. The parameters such as precision and F1 score produced a result synonymous with accuracy, as the dataset had a normal distribution and was not skewed.

| Models        | Precision | Recall | Accuracy | F1-score |
|---------------|-----------|--------|----------|----------|
| CNN           | 0.923     | 0.95   | 0.954    | 0.937    |
| KNN           | 0.564     | 0.962  | 0.302    | 0.736    |
| SVM           | 0.753     | 0.974  | 0.553    | 0.773    |
| Random Forest | 0.864     | 0.979  | 0.472    | 0.536    |

**Table 1.** Accuracy of Models on Test split 60:40

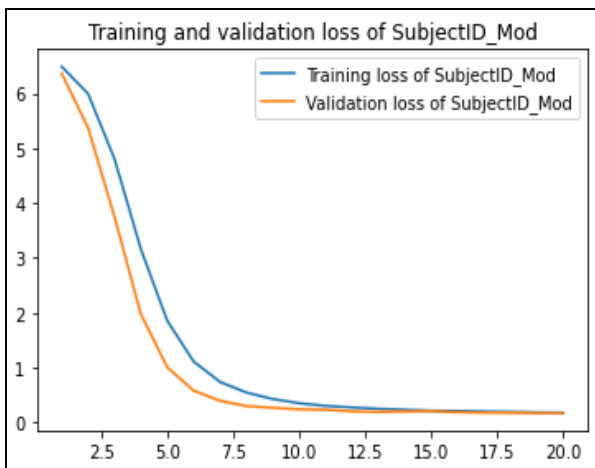
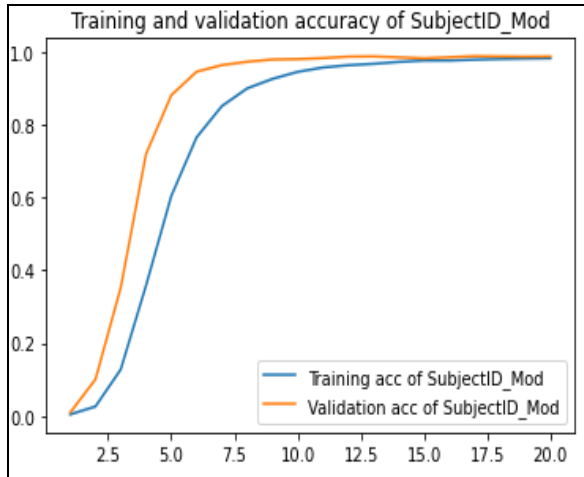
| Models        | Precision | Recall | Accuracy | F1-score |
|---------------|-----------|--------|----------|----------|
| CNN           | 0.924     | 0.949  | 0.976    | 0.936    |
| KNN           | 0.443     | 0.962  | 0.398    | 0.656    |
| SVM           | 0.723     | 0.973  | 0.398    | 0.781    |
| Random Forest | 0.227     | 0.981  | 0.424    | 0.428    |

**Table 2.** Accuracy of Models on Test split 90:5:5

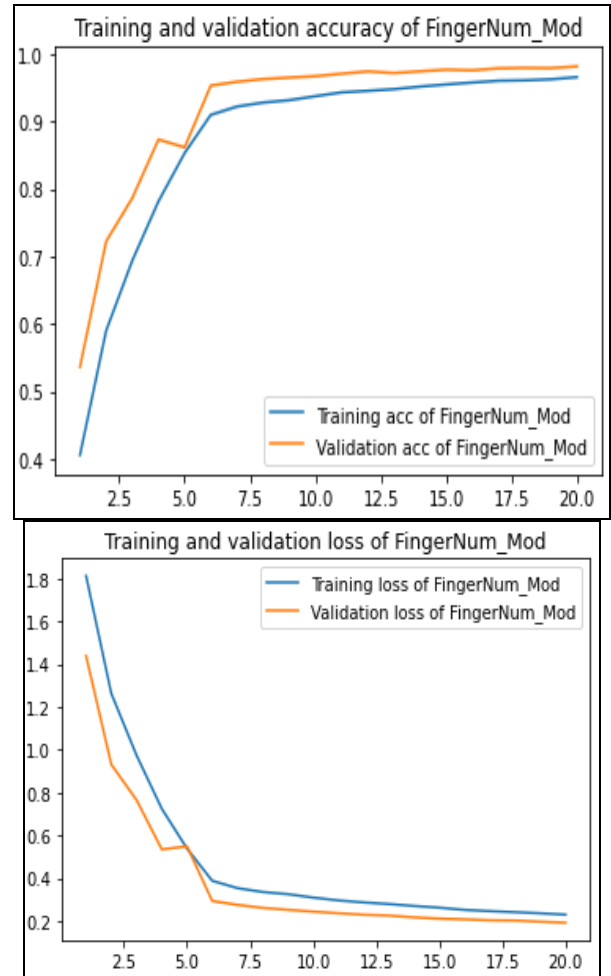
| Models        | Precision | Recall | Accuracy | F1-score |
|---------------|-----------|--------|----------|----------|
| CNN           | 0.93      | 0.953  | 0.962    | 0.941    |
| KNN           | 0.026     | 0.962  | 0.388    | 0.716    |
| SVM           | 0.763     | 0.98   | 0.603    | 0.812    |
| Random Forest | 0.346     | 0.989  | 0.301    | 0.573    |

**Table 3.** Accuracy of Models on Test split 90:10

By varying the train, validation, and test split ratios and through hyperparameter tuning, CNN is identified as the best possible solution in worst-case scenarios, where the images are tampered with using various data augmentation techniques. Each CNN model was trained for 20 epochs with a batch size of 64 images. Each training step fairly took 11 seconds when trained over GPU P100 accelerator iterating over 6000 images in total.



**Fig. 6a. & Fig 6b.** Plots describing the accuracy and validation of loss of SubjectID\_Mod CNN



**Fig. 7a. & Fig 7b.** Plots describing the accuracy and validation of loss of FingerNum\_Mod CNN

SubjectID\_Mod CNN model experiences a loss of 17.23% and FingerNum\_Mod CNN model experiences a loss of 19.25% as shown in Figure 6 and 7 respectively. The accuracy and loss were considered for the validation set of the dataset, thereby improving the fingerprint recognition performance during testing. SVM emerges as the second-best winner but it is highly sensitive to the quality of images, creating an impact in its accuracy. This work can be extended in the future as an end-to-end CNN-based recognition technique, that is built extending feature extraction utilising texture and pore identification. They can execute in short intervals of time, even over contactless and latent fingerprints.

## References

- [1] Darlow, L. N., Rosman, B. S.: "Fingerprint minutiae extraction using deep learning", IEEE International Joint Conference on Biometrics (IJCBI), pp. 22-30, 2017.
- [2] B. Zhou, C. Han, Y. Liu, T. Guo, and J. Qin, Fast minutiae extractor using neural network, Pattern Recognition, vol. 103, pp. 107273, 2020.
- [3] González, M., Gorziza, R. P., Mariotti, K. C., & Limberger, R. P. "Methodologies applied to fingerprint analysis", Journal of Forensic Sciences, 65(4), 1040-1048, 2020.
- [4] X. Jiang, W. Yau, and D. Youmaran, "Fingerprint Minutiae Matching Algorithm Based on Local and Global Features," Pattern Recognition, vol. 41, no. 8, pp. 2789-2807, 2008.
- [5] Hazarika, P., & Russell, D. A. "Advances in fingerprint analysis". Angewandte Chemie International Edition, 51(15), 3524-3531, 2012.
- [6] R. Cappelli, A. Erol, and D. Maio, "Fingerprint image retrieval with enhanced filter-based features," Pattern Recognition, vol. 45, no. 4, pp. 1472-1485, 2012.
- [7] K.H. Sun, Y.J. Liu, and J.J. Qin, "A fingerprint verification system based on decision fusion of minutiae and correlation matching," Pattern Recognition Letters, vol. 23, no. 1, pp. 11-18, 2002.
- [8] A.A. Goswami, U. Desai, A. Kumar, and M.N.S. Swamy, "Fingerprint matching using minutiae and texture features," Pattern Recognition, vol. 46, no. 7, pp. 1813-1822, 2013.
- [9] Moses Daluz, H. "Fundamentals of Fingerprint Analysis", Second Edition (2nd ed.). CRC Press, 2018.
- [10] Ai Takahashi, Yoshinori Koda, Koichi Ito, Takafumi Aoki, "Fingerprint Feature Extraction by Combining Texture, Minutiae, and Frequency Spectrum Using Multi-Task CNN," 27 Aug 2020.
- [11] Zhang, Z., Liu, S., & Liu, M. , A multi-task fully deep convolutional neural network for contactless fingerprint minutiae extraction. Pattern Recognition, 120, 108189, 2021.
- [12] Durú, B. M., Targino, J. M., & Lima, C. A. de M., Biometric Recognition Based on Fingerprint: A Comparative Study. 2017 Workshop of Computer Vision (WVC), 2017.
- [13] Prasad, P. S., et al. A Survey of Fingerprint Recognition Systems and Their Applications. Lecture Notes in Electrical Engineering, 500. Springer, Singapore, 2018.
- [14] Lovelesh Khard, Raju Baraskar, Uday Chaurasia . Survey on Fingerprint Recognition System using Image Preprocessing and Denoising. International Journal of Computer Applications. 155, pp. 39-43, 2016.
- [15] Shehu, Y. I., Ruiz-Garcia, A., Palade, V., & James, A., Sokoto Coventry Fingerprint Dataset, arXiv.org, <https://arxiv.org/abs/1807.10609>, 2021.
- [16] H. Guan et al., "Analysis, Comparison, and Assessment of Latent Fingerprint Image Preprocessing," 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Honolulu, HI, USA, pp. 628-635, 2017.
- [17] Chaurasia, O. P., An Approach to Fingerprint Image Pre-Processing. International Journal of Image, Graphics and Signal Processing, 4(6), 29-35, 2012.